RBUS6923

Data Analysis Assignment

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1 Question 1

1.1 Regression Estimation

Dep. Variable:	Y	**R-squared:**	0.631
Model:	OLS	**Adj. R-squared:**	0.484
Method:	Least Squares	**F-statistic:**	4.282
Date:	Sun, 05 Apr 2026	**Prob (F-statistic):**	0.0283
Time:	21:47:31	**Log-Likelihood:**	-54.698
No. Observations:	15	**AIC:**	119.4
Df Residuals:	10	**BIC:**	122.9
Df Model:	4		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
Intercept	-29.4689	23.568	-1.250	0.240	-81.982	23.044
X1	2.9472	2.125	1.387	0.196	-1.788	7.682
X2	4.6296	4.674	0.991	0.345	-5.784	15.043
X3	0.0219	0.012	1.754	0.110	-0.006	0.050
X4	0.0120	0.005	2.493	0.032	0.001	0.023

Omnibus:	0.991	**Durbin-Watson:**	1.231
Prob(Omnibus):	0.609	**Jarque-Bera (JB):**	0.245
Skew:	-0.311	**Prob(JB):**	0.885
Kurtosis:	3.073	**Cond. No.**	2.08e+04

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 2.08e+04. This might indicate that there are strong multicollinearity or other numerical problems.

Table 1: OLS regression results for equation (Q1a).

The estimated model is $\hat{Y}_i = -29.47 + 2.95X_{1,i} + 4.63X_{2,i} + 0.022X_{3,i} + 0.012X_{4,i}$. The model explains approximately 63.1% of the variation in Y ($R^2 = 0.631$, adjusted $R^2 = 0.484$). The overall model is statistically significant at the 5% level ($F = 4.282$, $p = 0.028$), indicating that the explanatory variables jointly contribute to explaining variation in the dependent variable. Of the individual coefficients, only X_4 is statistically significant at the 5% level ($t = 2.493$, $p = 0.032$); the remaining three variables are individually insignificant. The condition number is large (2.08×10^4), suggesting the

possibility of multicollinearity among the regressors.

1.2 Hypothesis Tests

The theory predicts specific values for each slope coefficient. Each hypothesis is tested separately using a two-sided t-test with $H_0\colon \beta_j = \beta_j^*$ and $H_1\colon \beta_j \neq \beta_j^*$, where $t = (\hat{\beta}_j - \beta_j^*)/\text{se}(\hat{\beta}_j)$ with 10 degrees of freedom.

	coef	std err	t	P>|t|	Conf. Int. Low	Conf. Int. Upp.
c0	2.947189	2.125020	0.445732	0.665286	-1.787651	7.682029

Table 2: Hypothesis test: $H_0\colon \beta_1 = 2.00$.

	coef	std err	t	P>|t|	Conf. Int. Low	Conf. Int. Upp.
c0	4.629577	4.673632	-0.614174	0.552816	-5.783924	15.043077

Table 3: Hypothesis test: $H_0\colon \beta_2 = 7.50$.

	coef	std err	t	P>|t|	Conf. Int. Low	Conf. Int. Upp.
c0	0.021912	0.012496	0.953259	0.362932	-0.005931	0.049755

Table 4: Hypothesis test: $H_0\colon \beta_3 = 0.01$.

	coef	std err	t	P>|t|	Conf. Int. Low	Conf. Int. Upp.
c0	0.012049	0.004834	-7.850897	0.000014	0.001278	0.022820

Table 5: Hypothesis test: $H_0\colon \beta_4 = 0.05$.

Testing each hypothesis at $\alpha = 0.05$:

(i) $H_0\colon \beta_1 = 2.00$. The test statistic is $t = (2.947 - 2.00)/2.125 = 0.446$ with $p = 0.665$. We fail to reject the null. The estimated coefficient is consistent with the theoretically predicted value of 2.00.

(ii) $H_0\colon \beta_2 = 7.50$. The test statistic is $t = (4.630 - 7.50)/4.674 = -0.614$ with $p = 0.553$. We fail to reject the null. The data does not provide sufficient evidence to distinguish $\hat{\beta}_2$ from the hypothesised value of 7.50.

(iii) $H_0\colon \beta_3 = 0.01$. The test statistic is $t = (0.022 - 0.01)/0.012 = 0.953$ with $p = 0.363$. We fail to reject the null. The estimated coefficient is consistent with the theoretically predicted value of 0.01.

(iv) $H_0\colon \beta_4 = 0.05$. The test statistic is $t = (0.012 - 0.05)/0.005 = -7.851$ with $p < 0.001$. We reject the null. The estimated coefficient $\hat{\beta}_4 = 0.012$ is statistically distinct from the hypothesised value of 0.05 at any conventional significance level.

In summary, the data supports the theoretical predictions for β_1, β_2, and β_3, but not for β_4.

1.3 Quadratic Functional Form

Dep. Variable:	Y	**R-squared:**	0.179
Model:	OLS	**Adj. R-squared:**	0.042
Method:	Least Squares	**F-statistic:**	1.310
Date:	Sun, 05 Apr 2026	**Prob (F-statistic):**	0.306
Time:	21:47:31	**Log-Likelihood:**	-60.702
No. Observations:	15	**AIC:**	127.4
Df Residuals:	12	**BIC:**	129.5
Df Model:	2		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
Intercept	68.6094	44.778	1.532	0.151	-28.954	166.173
X4	-0.0147	0.049	-0.303	0.767	-0.121	0.091
X4_sq	5.506e-06	1.16e-05	0.475	0.643	-1.98e-05	3.08e-05

Omnibus:	1.508	**Durbin-Watson:**	1.017
Prob(Omnibus):	0.470	**Jarque-Bera (JB):**	1.011
Skew:	-0.345	**Prob(JB):**	0.603
Kurtosis:	1.931	**Cond. No.**	5.70e+07

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 5.7e+07. This might indicate that there are strong multicollinearity or other numerical problems.

Table 6: Quadratic model results for equation (Q1c).

The quadratic specification $\hat{Y}_i = 68.61 - 0.015X_{4,i} + 5.506 \times 10^{-6}X_{4,i}^2$ is poorly specified. Neither the linear term ($p = 0.767$) nor the quadratic term ($p = 0.643$) is individually significant, and the overall model fails to achieve global significance ($F = 1.310$, $p = 0.306$). The R^2 of 0.179 represents a substantial deterioration from the 0.631 achieved in the four-variable linear model in table 1. These results provide no evidence that the relationship between Y and X_4 is non-linear. The linear functional form estimated in Part (a) is the preferred specification.

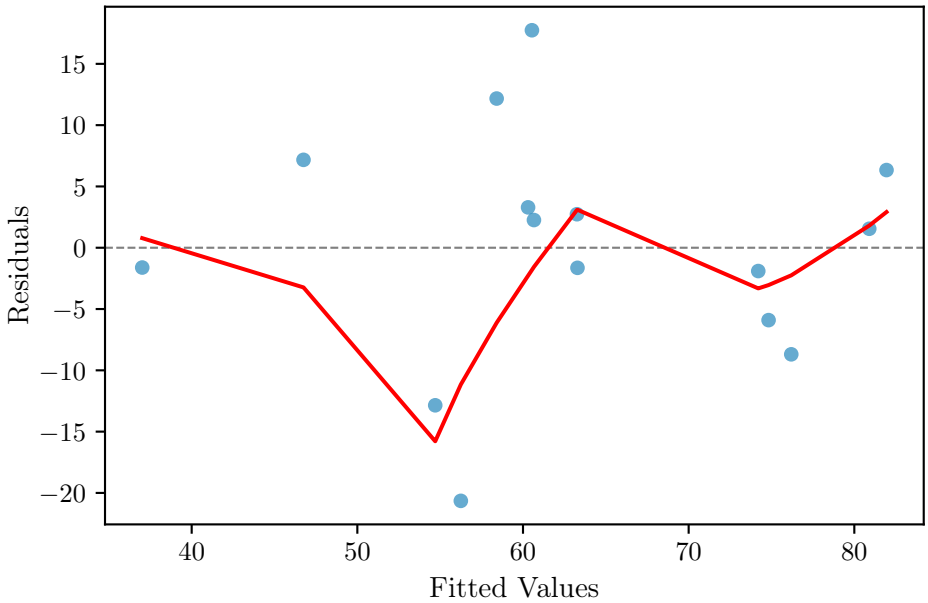

Figure 1: Residuals versus fitted values for the linear model (Q1a). The loess curve shows no systematic non-linear pattern, confirming the adequacy of the linear functional form.

2 Question 2

Dep. Variable:	V	**R-squared:**	0.773
Model:	OLS	**Adj. R-squared:**	0.767
Method:	Least Squares	**F-statistic:**	120.5
Date:	Sun, 05 Apr 2026	**Prob (F-statistic):**	6.36e-76
Time:	21:47:31	**Log-Likelihood:**	-2352.6
No. Observations:	255	**AIC:**	4721.
Df Residuals:	247	**BIC:**	4750.
Df Model:	7		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
Intercept	274.9603	314.896	0.873	0.383	-345.264	895.184
ABV	2.1910	0.150	14.561	0.000	1.895	2.487
AE	8.1527	0.634	12.852	0.000	6.903	9.402
NECE	-3.0712	1.075	-2.856	0.005	-5.189	-0.953
POLLUTE	-912.2288	465.298	-1.961	0.051	-1828.688	4.230
NECE:POLLUTE	9.0744	1.402	6.474	0.000	6.314	11.835
ECE	0.8602	0.798	1.079	0.282	-0.711	2.431
ECE:POLLUTE	-3.4427	1.106	-3.112	0.002	-5.622	-1.264

Omnibus:	113.453	**Durbin-Watson:**	0.962
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	1013.172
Skew:	1.523	**Prob(JB):**	9.83e-221
Kurtosis:	12.278	**Cond. No.**	8.24e+03

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 8.24e+03. This might indicate that there are strong multicollinearity or other numerical problems.

Table 7: Market valuation model results for equation (Q2).

	coef	std err	t	P>|t|	Conf. Int. Low	Conf. Int. Upp.
c0	-2.582514	0.805199	-3.207301	0.001517	-4.168445	-0.996583

Table 8: t-test for $\beta_5 + \beta_6 = 0$.

F-statistic	p-value	df_num	df_denom
10.286779	0.001517	1.000000	247.000000

Table 9: F-test for $\beta_5 + \beta_6 = 0$.

The model achieves an R^2 of 0.773 and is highly significant overall ($F = 120.5$, $p < 0.001$). The three predictions of Clarkson, Li, and Richardson [1] are evaluated as follows.

Prediction 1 posits that ECEs of low polluting firms are valued as assets (NPV > 0). For low polluters (POLLUTE $= 0$), the market valuation of ECE is captured by β_5 alone. The estimated coefficient is $\hat{\beta}_5 = 0.860$ ($t = 1.079$, $p = 0.282$), which is positive but statistically indistinguishable from zero. The data therefore does not provide significant support for the prediction that the market views low-polluter ECE as a value-creating asset.

Prediction 2 posits that ECEs of high polluting firms have zero net present value (NPV $= 0$). For high polluters (POLLUTE $= 1$), the market valuation of ECE is $\beta_5 + \beta_6$. The linear combination yields $0.860 + (-3.443) = -2.583$, which is significantly different from zero ($t = -3.207$, $p = 0.002$; $F = 10.29$, $p = 0.002$). Because the net coefficient is significantly negative rather than zero, the data contradicts the prediction. The market appears to penalise environmental capital expenditure by high polluting firms, suggesting the market perceives such spending as value-destroying rather than merely having zero net benefit.

Prediction 3 posits that high polluting firms have unbooked environmental liabilities, implying a lower market value all else equal. This is tested via the POLLUTE indicator coefficient β_7. The estimate is $\hat{\beta}_7 = -912.23$ ($t = -1.961$, $p = 0.051$), which is marginally significant at the 10% level but falls just short of the conventional 5% threshold. This provides weak support for the prediction that high polluting firms carry implicit environmental liabilities not captured by the balance sheet.

3 Question 3

3.1 OLS Regression

Dep. Variable:	V	**R-squared:**	0.511
Model:	OLS	**Adj. R-squared:**	0.511
Method:	Least Squares	**F-statistic:**	2572.
Date:	Sun, 05 Apr 2026	**Prob (F-statistic):**	0.00
Time:	21:47:33	**Log-Likelihood:**	-1.3834e+05
No. Observations:	12289	**AIC:**	2.767e+05
Df Residuals:	12283	**BIC:**	2.767e+05
Df Model:	5		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
Intercept	-235.1720	762.963	-0.308	0.758	-1730.700	1260.356
BV	1.2547	0.014	92.091	0.000	1.228	1.281
Earn	0.8111	0.183	4.431	0.000	0.452	1.170
CSR_Perf	3008.5897	1394.684	2.157	0.031	274.791	5742.389
CSR_Info	-744.9244	2809.924	-0.265	0.791	-6252.816	4762.967
Assure	2061.7573	499.747	4.126	0.000	1082.174	3041.340

Omnibus:	39708.062	**Durbin-Watson:**	1.142
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	5395222957.630
Skew:	53.890	**Prob(JB):**	0.00
Kurtosis:	3247.237	**Cond. No.**	3.60e+05

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 3.6e+05. This might indicate that there are strong multicollinearity or other numerical problems.

Table 10: OLS regression results for equation (Q3).

Under a naive OLS specification with 12,289 firm-year observations, the assurance decision is highly valuation relevant. The coefficient on Assure is $\hat{\delta}_5 = 2061.76$ ($t = 4.126$, $p < 0.001$), indicating that firms with assured CSR reports are associated with market values approximately \$2,062 million higher, controlling for book value, earnings, CSR performance, and CSR disclosure. Book value ($t = 92.09$) and earnings ($t = 4.43$) are also highly significant. CSR performance is marginally significant ($p = 0.031$), while CSR

disclosure is not ($p = 0.791$).

3.2 Clustered Standard Errors

Dep. Variable:	V	**R-squared:**	0.511
Model:	OLS	**Adj. R-squared:**	0.511
Method:	Least Squares	**F-statistic:**	1304.
Date:	Sun, 05 Apr 2026	**Prob (F-statistic):**	2.00e-26
Time:	21:47:33	**Log-Likelihood:**	-1.3834e+05
No. Observations:	12289	**AIC:**	2.767e+05
Df Residuals:	12283	**BIC:**	2.767e+05
Df Model:	5		
Covariance Type:	cluster		

	coef	**std err**	**z**	**P> |z|**	**[0.025**	**0.975]**
Intercept	-235.1720	1568.096	-0.150	0.881	-3308.583	2838.239
BV	1.2547	0.022	56.109	0.000	1.211	1.299
Earn	0.8111	0.470	1.725	0.084	-0.110	1.732
CSR_Perf	3008.5897	2631.781	1.143	0.253	-2149.607	8166.786
CSR_Info	-744.9244	5672.402	-0.131	0.896	-1.19e+04	1.04e+04
Assure	2061.7573	1324.295	1.557	0.120	-533.814	4657.328

Omnibus:	39708.062	**Durbin-Watson:**	1.142
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	5395222957.630
Skew:	53.890	**Prob(JB):**	0.00
Kurtosis:	3247.237	**Cond. No.**	3.60e+05

Notes:
[1] Standard Errors are robust to cluster correlation (cluster)
[2] The condition number is large, 3.6e+05. This might indicate that there are strong multicollinearity or other numerical problems.

Table 11: Regression with standard errors clustered by industry.

Clustering standard errors by industry leaves all point estimates identical to Part (a) but substantially inflates the standard errors. The key changes are as follows. The standard error for Assure increases from 499.75 to 1324.30, rendering the coefficient statistically insignificant ($z = 1.557$, $p = 0.120$). Earnings shifts from significant ($p < 0.001$) to marginally significant ($p = 0.084$). CSR performance becomes insignificant ($p = 0.253$). Only book value retains strong significance ($z = 56.11$, $p < 0.001$).

These differences are expected. Naive OLS assumes that observations are independent and identically distributed. When firms within the same industry share common unobserved shocks, the residuals are correlated within clusters. The naive standard errors understate the true sampling variability, artificially inflating t-statistics. Clustering corrects for this intra-industry dependence by relaxing the independence assumption, producing conservative and more reliable inference.

3.3 Country Fixed Effects

Dep. Variable:	V	**R-squared:**	0.513
Model:	OLS	**Adj. R-squared:**	0.512
Method:	Least Squares	**F-statistic:**	977.0
Date:	Sun, 05 Apr 2026	**Prob (F-statistic):**	1.33e-27
Time:	21:47:33	**Log-Likelihood:**	-1.3832e+05
No. Observations:	12289	**AIC:**	2.767e+05
Df Residuals:	12272	**BIC:**	2.768e+05
Df Model:	16		
Covariance Type:	cluster		

	coef	**std err**	**z**	**P> |z|**	**[0.025**	**0.975]**
Intercept	552.1592	1842.330	0.300	0.764	-3058.741	4163.059
BV	1.2509	0.022	57.304	0.000	1.208	1.294
Earn	0.8236	0.467	1.762	0.078	-0.093	1.740
CSR_Perf	3325.2289	2897.841	1.147	0.251	-2354.436	9004.893
CSR_Info	-2217.6456	6055.372	-0.366	0.714	-1.41e+04	9650.665
Assure	1979.7239	1398.613	1.415	0.157	-761.508	4720.955
country2	-454.7398	624.577	-0.728	0.467	-1678.889	769.409
country3	-868.7610	1002.310	-0.867	0.386	-2833.253	1095.730
country4	-696.5450	536.803	-1.298	0.194	-1748.659	355.569
country5	-800.4621	770.835	-1.038	0.299	-2311.271	710.347
country6	4459.1666	4636.582	0.962	0.336	-4628.367	1.35e+04
country7	-239.1246	584.697	-0.409	0.683	-1385.110	906.861
country8	-1503.6913	585.386	-2.569	0.010	-2651.026	-356.356
country9	-1474.7326	683.749	-2.157	0.031	-2814.857	-134.609
country10	-723.6646	1030.144	-0.702	0.482	-2742.709	1295.380
country11	1895.7738	1354.566	1.400	0.162	-759.126	4550.674
country12	-609.6965	1628.668	-0.374	0.708	-3801.827	2582.434

Omnibus:	39692.402	**Durbin-Watson:**	1.141
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	5377274652.753
Skew:	53.830	**Prob(JB):**	0.00
Kurtosis:	3241.834	**Cond. No.**	3.66e+05

Notes:
[1] Standard Errors are robust to cluster correlation (cluster)
[2] The condition number is large, 3.66e+05. This might indicate that there are strong multicollinearity or other numerical problems.

Table 12: Regression with country fixed effects and clustered standard errors.

Adding country fixed effects to absorb unobserved, time-invariant national heterogeneity produces only a marginal change in model fit (R^2 increases from 0.511 to 0.513). The Assure coefficient shifts slightly to 1979.72 ($z = 1.415$, $p = 0.157$) and remains statistically insignificant, corroborating the conclusion from Part (b). Two country dummies are individually significant (country 8 and country 9), suggesting systematic cross-country valuation differences. The core variables of interest are largely unaffected by the inclusion of fixed effects, indicating that the insignificance of the assurance variable under clustered standard errors is not an artefact of omitted country-level confounders.

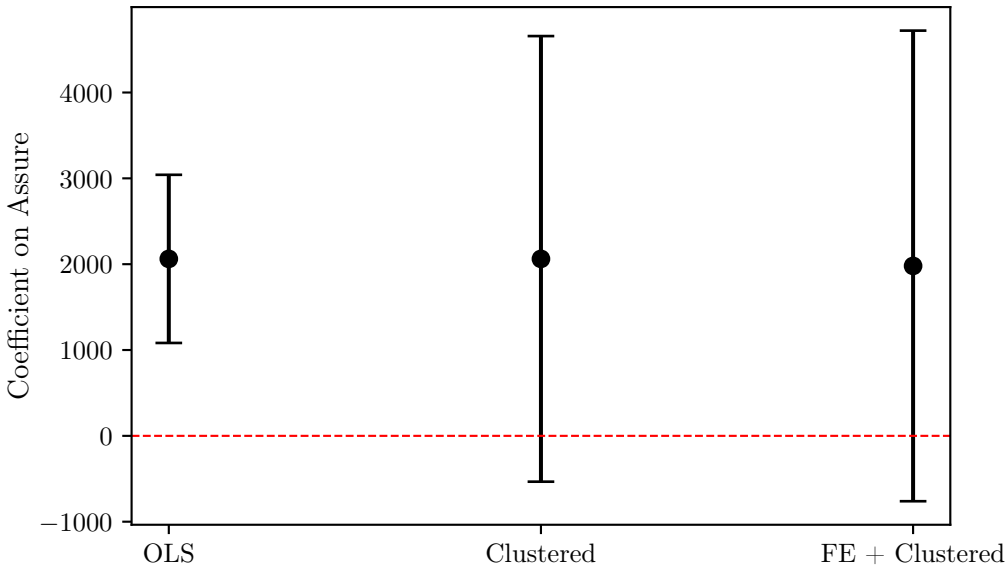

Figure 2: CSR assurance coefficient estimates with 95% confidence intervals across specifications. The interval includes zero under clustered and fixed-effect specifications.

4 Question 4

4.1 January Effect

Dep. Variable:	rp_sml_rf	**R-squared:**	0.563
Model:	OLS	**Adj. R-squared:**	0.560
Method:	Least Squares	**F-statistic:**	206.5
Date:	Sun, 05 Apr 2026	**Prob (F-statistic):**	2.23e-58
Time:	21:47:33	**Log-Likelihood:**	317.52
No. Observations:	324	**AIC:**	-629.0
Df Residuals:	321	**BIC:**	-617.7
Df Model:	2		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
Intercept	0.0535	0.005	9.890	0.000	0.043	0.064
d_jan	0.0528	0.019	2.849	0.005	0.016	0.089
rm_ew_rf	1.7129	0.088	19.527	0.000	1.540	1.885

Omnibus:	218.432	**Durbin-Watson:**	1.599
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	2897.666
Skew:	2.605	**Prob(JB):**	0.00
Kurtosis:	16.693	**Cond. No.**	17.4

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Table 13: January effect regression for small stocks.

Dep. Variable:	rp_lrg_rf	**R-squared:**	0.573
Model:	OLS	**Adj. R-squared:**	0.570
Method:	Least Squares	**F-statistic:**	215.0
Date:	Sun, 05 Apr 2026	**Prob (F-statistic):**	5.79e-60
Time:	21:47:33	**Log-Likelihood:**	634.92
No. Observations:	324	**AIC:**	-1264.
Df Residuals:	321	**BIC:**	-1252.
Df Model:	2		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
Intercept	-0.0057	0.002	-2.792	0.006	-0.010	-0.002
d_jan	-0.0015	0.007	-0.222	0.824	-0.015	0.012
rm_ew_rf	0.6772	0.033	20.561	0.000	0.612	0.742

Omnibus:	33.464	**Durbin-Watson:**	1.886
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	155.655
Skew:	-0.203	**Prob(JB):**	1.59e-34
Kurtosis:	6.371	**Cond. No.**	17.4

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Table 14: January effect regression for large stocks.

The model $R_{p,t} - R_{f,t} = \alpha_1 + \alpha_2 D_{\text{Jan},t} + \beta_p(R_{M,t} - R_{f,t}) + \varepsilon_t$ is estimated separately for each portfolio using the equal-weighted market return. For the small stock portfolio, the January dummy coefficient is $\hat{\alpha}_2 = 0.0528$ ($t = 2.849$, $p = 0.005$), indicating a statistically significant incremental abnormal return of approximately 5.3 percentage points in January. The intercept is also significant ($\hat{\alpha}_1 = 0.0535$, $p < 0.001$), reflecting a positive average abnormal return in non-January months. The portfolio beta is $\hat{\beta}_p = 1.713$, consistent with the higher systematic risk of small capitalisation stocks.

For the large stock portfolio, the January dummy is $\hat{\alpha}_2 = -0.0015$ ($t = -0.222$, $p = 0.824$), which is statistically insignificant. There is no evidence of a January effect for large stocks. The intercept is significantly negative ($\hat{\alpha}_1 = -0.0057$, $p = 0.006$), indicating a small negative abnormal return in non-January months. The portfolio beta is $\hat{\beta}_p = 0.677$, reflecting the lower systematic risk of large stocks relative to the equal-weighted market.

Dep. Variable:	rp_rf	**R-squared:**	0.619
Model:	OLS	**Adj. R-squared:**	0.616
Method:	Least Squares	**F-statistic:**	209.0
Date:	Sun, 05 Apr 2026	**Prob (F-statistic):**	4.44e-132
Time:	21:47:33	**Log-Likelihood:**	816.90
No. Observations:	648	**AIC:**	-1622.
Df Residuals:	642	**BIC:**	-1595.
Df Model:	5		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
Intercept	-0.0057	0.004	-1.388	0.166	-0.014	0.002
d_jan	-0.0015	0.014	-0.111	0.912	-0.029	0.026
d_sml	0.0592	0.006	10.240	0.000	0.048	0.071
d_jan:d_sml	0.0543	0.020	2.745	0.006	0.015	0.093
rm_ew_rf	0.6772	0.066	10.221	0.000	0.547	0.807
rm_ew_rf:d_sml	1.0357	0.094	11.054	0.000	0.852	1.220

Omnibus:	489.635	**Durbin-Watson:**	1.636
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	15064.100
Skew:	3.010	**Prob(JB):**	0.00
Kurtosis:	25.840	**Cond. No.**	45.5

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Table 15: Pooled regression for small firm and differential January effects.

To formally test for a small firm effect and a differential January effect, the data for both portfolios are pooled and the following model is estimated:

$$\begin{aligned} R_{p,t} - R_{f,t} = \gamma_0 + \gamma_1 D_{\text{Jan},t} + \gamma_2 D_{\text{Sml},t} + \gamma_3 (D_{\text{Jan},t} \times D_{\text{Sml},t}) \\ + \beta_1 (R_{M,t} - R_{f,t}) + \beta_2 \big((R_{M,t} - R_{f,t}) \times D_{\text{Sml},t}\big) + \varepsilon_t \end{aligned}$$

The coefficient $\hat{\gamma}_2 = 0.0592$ ($t = 10.24$, $p < 0.001$) confirms a significant small firm effect: small stocks earn approximately 5.9 percentage points higher abnormal returns per month than large stocks, after controlling for market risk. The interaction term $\hat{\gamma}_3 = 0.0543$ ($t = 2.745$, $p = 0.006$) confirms that the January effect is significantly larger for small stocks than for large stocks. The market risk interaction $\hat{\beta}_2 = 1.036$ ($p < 0.001$) reflects the higher beta of the small stock portfolio.

4.2 July Effect

Dep. Variable:	rp_sml_rf	**R-squared:**	0.236
Model:	OLS	**Adj. R-squared:**	0.228
Method:	Least Squares	**F-statistic:**	32.88
Date:	Sun, 05 Apr 2026	**Prob (F-statistic):**	1.49e-18
Time:	21:47:33	**Log-Likelihood:**	227.07
No. Observations:	324	**AIC:**	-446.1
Df Residuals:	320	**BIC:**	-431.0
Df Model:	3		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
d_oth	0.0616	0.007	8.368	0.000	0.047	0.076
d_jan	0.1589	0.023	6.792	0.000	0.113	0.205
d_jul	0.1728	0.023	7.415	0.000	0.127	0.219
rm_vw_rf	0.9436	0.126	7.485	0.000	0.696	1.192

Omnibus:	145.761	**Durbin-Watson:**	1.487
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	857.256
Skew:	1.786	**Prob(JB):**	7.07e-187
Kurtosis:	10.123	**Cond. No.**	17.2

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Table 16: January and July effect regression for small stocks.

Dep. Variable:	rp_lrg_rf	**R-squared:**	0.947
Model:	OLS	**Adj. R-squared:**	0.947
Method:	Least Squares	**F-statistic:**	1914.
Date:	Sun, 05 Apr 2026	**Prob (F-statistic):**	5.82e-204
Time:	21:47:33	**Log-Likelihood:**	973.72
No. Observations:	324	**AIC:**	-1939.
Df Residuals:	320	**BIC:**	-1924.
Df Model:	3		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
d_oth	-0.0009	0.001	-1.277	0.203	-0.002	0.001
d_jan	0.0017	0.002	0.744	0.458	-0.003	0.006
d_jul	0.0030	0.002	1.287	0.199	-0.002	0.008
rm_vw_rf	0.9459	0.013	75.179	0.000	0.921	0.971

Omnibus:	11.642	**Durbin-Watson:**	1.567
Prob(Omnibus):	0.003	**Jarque-Bera (JB):**	22.449
Skew:	-0.124	**Prob(JB):**	1.33e-05
Kurtosis:	4.266	**Cond. No.**	17.2

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Table 17: January and July effect regression for large stocks.

To test for both a January and a July effect, the intercept is suppressed and three mutually exclusive dummy variables are included:

$$R_{p,t} - R_{f,t} = \delta_1 D_{\text{Jan},t} + \delta_2 D_{\text{Jul},t} + \delta_3 D_{\text{Oth},t} + \beta_p (R_{M,t} - R_{f,t}) + \varepsilon_t$$

where $D_{\text{Jan},t}$, $D_{\text{Jul},t}$, and $D_{\text{Oth},t}$ are indicator variables for January, July, and all other months respectively. The value-weighted market return is used for $R_{M,t}$. Each coefficient directly measures the average abnormal return in the corresponding period.

For small stocks, all three period coefficients are highly significant: January ($\hat{\delta}_1 = 0.1589$, $p < 0.001$), July ($\hat{\delta}_2 = 0.1728$, $p < 0.001$), and other months ($\hat{\delta}_3 = 0.0616$, $p < 0.001$). The July abnormal return of 17.3% exceeds the January abnormal return of 15.9%, both of which are substantially larger than the 6.2% earned in other months. This is consistent with tax-loss selling around the Australian 30 June fiscal year-end driving a July rebound effect in small capitalisation stocks.

For large stocks, none of the three period coefficients are individually significant: January ($\hat{\delta}_1 = 0.0017$, $p = 0.458$), July ($\hat{\delta}_2 = 0.0030$, $p = 0.199$), and other months ($\hat{\delta}_3 = -0.0009$, $p = 0.203$). The large stock portfolio exhibits no seasonal abnormal return patterns after controlling for market risk.

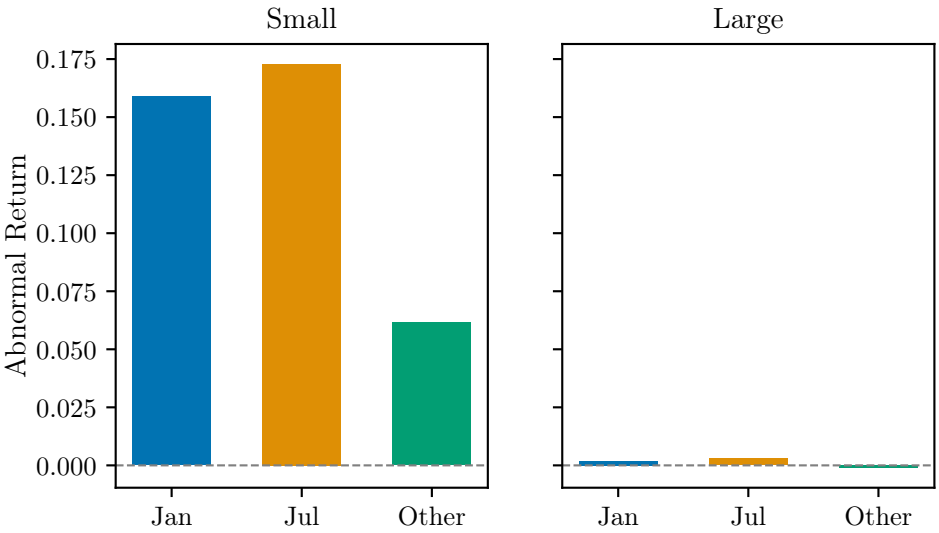

Figure 3: Seasonal abnormal returns for small and large stock portfolios. Small stocks exhibit significant January and July effects; large stocks show no seasonal pattern.

4.3 Pre and Post Crash Comparison

Dep. Variable:	vw_mkt	**R-squared:**	0.026
Model:	OLS	**Adj. R-squared:**	0.017
Method:	Least Squares	**F-statistic:**	2.786
Date:	Sun, 05 Apr 2026	**Prob (F-statistic):**	0.0409
Time:	21:47:33	**Log-Likelihood:**	506.51
No. Observations:	312	**AIC:**	-1005.
Df Residuals:	308	**BIC:**	-990.0
Df Model:	3		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
d1_jan	0.0490	0.013	3.679	0.000	0.023	0.075
d1_oth	0.0142	0.004	3.523	0.000	0.006	0.022
d2_jan	0.0083	0.013	0.625	0.532	-0.018	0.035
d2_oth	0.0093	0.004	2.315	0.021	0.001	0.017

Omnibus:	13.506	**Durbin-Watson:**	1.758
Prob(Omnibus):	0.001	**Jarque-Bera (JB):**	21.045
Skew:	-0.290	**Prob(JB):**	2.69e-05
Kurtosis:	4.132	**Cond. No.**	3.32

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Table 18: Sub-period January effect regression.

	coef	std err	t	P>|t|	Conf. Int. Low	Conf. Int. Upp.
c0	0.040673	0.018839	2.158952	0.031626	0.003603	0.077743

Table 19: Test for difference in January returns across sub-periods.

	coef	std err	t	P>|t|	Conf. Int. Low	Conf. Int. Upp.
c0	0.004853	0.005680	0.854440	0.393526	-0.006324	0.016030

Table 20: Test for difference in non-January returns across sub-periods.

To test whether the January effect changed after the October 1987 crash, the year 1987 is excluded and the sample is partitioned into Period 1 (January 1974 to December 1986)

and Period 2 (January 1988 to December 2000). The intercept is suppressed to obtain direct estimates of mean raw returns for the value-weighted market index:

$$R_{M,t} = \lambda_1 D_{1,\text{Jan},t} + \lambda_2 D_{1,\text{Oth},t} + \lambda_3 D_{2,\text{Jan},t} + \lambda_4 D_{2,\text{Oth},t} + \varepsilon_t$$

where $D_{1,\text{Jan},t}$ and $D_{1,\text{Oth},t}$ are indicators for January and non-January months in Period 1, and $D_{2,\text{Jan},t}$ and $D_{2,\text{Oth},t}$ are the corresponding indicators for Period 2. Each coefficient directly measures the average raw return in that period-month category.

The estimates are: (i) average January return in Period 1: $\hat{\lambda}_1 = 4.90\%$ ($t = 3.679$, $p < 0.001$); (ii) average non-January return in Period 1: $\hat{\lambda}_2 = 1.42\%$ ($t = 3.523$, $p < 0.001$); (iii) average January return in Period 2: $\hat{\lambda}_3 = 0.83\%$ ($t = 0.625$, $p = 0.532$); and (iv) average non-January return in Period 2: $\hat{\lambda}_4 = 0.93\%$ ($t = 2.315$, $p = 0.021$).

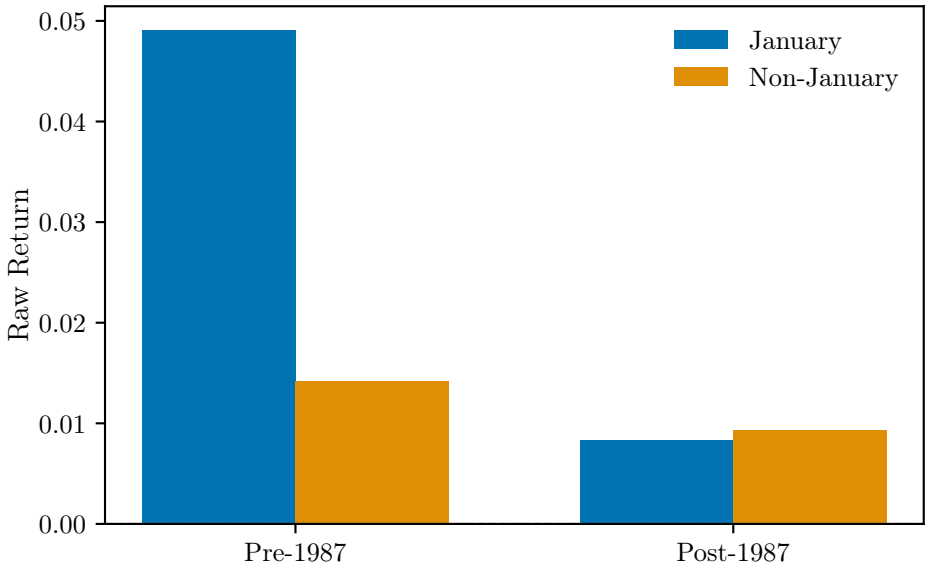

Figure 4: Average monthly returns by sub-period. The January effect diminished substantially after the 1987 crash while non-January returns remained stable.

To test for structural change:

The first hypothesis tests whether the January effect differs across sub-periods: $H_0\colon \lambda_1 = \lambda_3$. The difference is $\hat{\lambda}_1 - \hat{\lambda}_3 = 4.90\% - 0.83\% = 4.07\%$ with $t = 2.159$ and $p = 0.032$. We reject the null at the 5% level. The January effect was significantly larger in the pre-crash period, declining from a substantial 4.9% average return to an insignificant

0.83% in the post-crash period. This is consistent with the hypothesis that market microstructure changes and increased institutional participation following the crash reduced the magnitude of the January anomaly.

The second hypothesis tests whether non-January returns differ across sub-periods: $H_0\colon \lambda_2 = \lambda_4$. The difference is $\hat{\lambda}_2 - \hat{\lambda}_4 = 1.42\% - 0.93\% = 0.49\%$ with $t = 0.854$ and $p = 0.394$. We fail to reject the null. The baseline non-January market returns remained statistically invariant across the two periods, indicating that the structural change was specific to the January seasonal pattern rather than reflecting a general shift in market returns.

References

[1] P. M. Clarkson, Y. Li, and G. D. Richardson. The market valuation of environmental capital expenditures by pulp and paper companies. 79(2):329–353.